CONFIDENTIAL INVESTOR & TECHNICAL BRIEF

H~M Trading Institute

Investor & Technical Brief

AI-assisted trading research, built for African retail investors — subscription SaaS, Nigerian entity, live engineering.

Prepared: August 2026

Status: Pre-launch — engineering complete through core platform, admin, billing and monitoring phases

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Executive Summary

H~M Trading Institute is a multi-user subscription platform that gives retail investors in Nigeria and across Africa institutional-grade AI trading research — the same class of agentic analysis, backtesting and market scanning used by professional desks — at a price an individual can afford. Users bring their own DeepSeek API key (a BYOK model); the platform provides everything around it: signup, secure key storage, run queueing, a 462-alpha research engine, swarm orchestration, results rendering, and NGN billing via Paystack.

The BYOK design is the core of the unit economics: the customer pays the LLM provider directly, so the platform's revenue is subscription with structurally near-zero variable cost per user. Every subscription tier is unlimited-run; gross margin is therefore high by construction, and the pricing (₦20,000 / ₦35,000 / ₦75,000 per month ≈ US$12–47) sits far below the cost of Western equivalents while using local payment rails.

We are pre-launch and seeking investment to do three things: (1) take the finished platform to market with a real acquisition budget, (2) complete the compliance and entity layer a Nigerian financial-research business must have, and (3) expand the product surface (public API, mobile, international billing) that turns a local launch into a pan-African company. The engineering is done — the capital is for the company, not the code.

This document is written so a technical partner can explain the system end-to-end: market problem, infrastructure rationale, architecture, user/admin design flows, functional and non-functional requirements, data and security model, and the phased delivery record (Sections 2–13).

1 · Why We Need Investors

The platform is engineered and shipped through seven delivery phases; what it has never had is capital. Three concrete gaps require it:

1. Distribution. A research product with zero marketing budget cannot be discovered. Retail trading in Nigeria is won on the ground — creator channels, trading communities, referral loops — and that costs money before it compounds.
2. Compliance and entity cost. Operating a paid financial-research service in Nigeria requires legal review, registrant identity (physical/mailing address, registered business name), data-protection posture, and support infrastructure. These are fixed, non-optional costs.
3. Working capital and expansion. Paystack settlement float, compute scaling at launch, the public API, mobile views, and international (Stripe) billing all need cash runway — not more engineering.

1.1 Use of Funds (indicative allocation)

Category	Share	What it buys
Go-to-market & acquisition	40%	Creator partnerships, trading-community campaigns, referral incentives, content engine
Compliance, legal & entity	20%	Registrant block, ToS/Privacy enforceability, data-protection review, support mailbox & tooling
Product expansion	25%	Public API, mobile-optimized views, international billing (Stripe), admin 2FA, advanced analytics
Working capital & runway	15%	Paystack settlement float, infra at launch, contingency

Exact raise size to be agreed with the founders; allocation above is indicative and negotiation-ready.

1.2 The Return Case

- Recurring, prepaid revenue: monthly NGN subscriptions collected by Paystack, Nigeria's dominant payment processor.
- Structurally high margins: customers pay DeepSeek directly for tokens — no LLM cost line, no chargeback exposure on model usage.

- Defensible technology: a vendored 462-alpha research engine, 8 backtest engines, 24 data sources, ReAct agents and 30 swarm presets — packaged as a simple web product.
- Sitting on a growth wave: African retail investment adoption is rising sharply (local brokerage and fractional-investment apps have grown to millions of users); no localized AI research product currently owns this price point.
- Exit optionality: subscription SaaS with clean unit economics is acquirable by brokers, fintechs and data platforms seeking engaged retail traders.

2 · The Problem

Retail investors in Nigeria and across emerging markets face a research gap. Professional-grade trading research — quantitative alpha screening, multi-asset backtesting, news-and-data synthesis — is either locked inside institutions, priced in dollars for Western salaries, or buried inside fragmented tools that assume US/EU infrastructure: USD pricing, foreign card rails, and no local payment or support context.

Meanwhile the tools that do exist for African retail investors (fractional brokerage, savings/investment apps) are execution platforms — they let users buy, but they do not help users think. The result is a market of self-directed traders making decisions on sentiment, with no accessible research layer, at exactly the moment AI made such a layer cheap to operate.

Two structural facts make this solvable now: (1) frontier LLMs are available per-token to anyone with a card, and (2) a BYOK model lets a small company offer unlimited AI research without bearing model cost — the customer pays for intelligence, the platform sells the product around it.

3 · The Solution

3.1 Product

A web application where a user signs up, connects their own DeepSeek API key, and launches AI trading research agents: natural-language prompts become runs that scan markets, backtest strategies, and return rendered artifacts (charts, tables, reports) — with live progress streaming. Three subscription tiers gate capability depth, not usage: unlimited runs on every tier.

Tier	Price (NGN)	≈ USD	Capabilities
Starter	₦20,000/mo	≈ $12	Single-agent research runs, unlimited
Pro	₦35,000/mo	≈ $22	Starter + Swarm

			orchestration (30 multi-agent presets)
Premium	₦75,000/mo	≈ $47	Pro + attachment uploads (CSV/XLSX/JSON) for custom universe analysis

USD figures approximate at ≈₦1,600/USD. Users pay DeepSeek directly for tokens (typical retail usage is a small monthly LLM bill); the platform charges a flat, prepaid subscription for the research product around it. A soft safety net of 30 runs/hour/user prevents abuse without creating a business quota.

3.2 Why BYOK (Bring Your Own Key)

- Zero LLM margin risk: the platform never proxies or marks up model calls; there is no COGS line that scales with usage.
- Trust: users' keys are encrypted at rest in Supabase Vault; the platform can never see or spend them.
- Unlimited positioning: since tokens are the customer's, 'unlimited runs' is a true statement, not a marketing caveat.
- Simplified compliance: no handling of customer LLM content through our own payment flow.

4 · Why This Infrastructure

The architecture was chosen deliberately for a two-person, capital-efficient company serving paying customers in Nigeria:

- No traditional backend server. Supabase (PostgreSQL 17 + Auth + Storage + Vault + Realtime + Edge Functions) is the entire backend. Row-Level Security and SECURITY DEFINER RPCs enforce access in the database itself — the most audit-friendly security model available, and zero servers to patch.
- Paystack only, in NGN. The company is a Nigerian entity; Stripe cannot onboard Nigerian businesses. Hosted checkout, recurring plans, and signed webhooks (HMAC-SHA512) handle billing end-to-end.
- A Python polling worker instead of a job queue. Runs are claimed with FOR UPDATE SKIP LOCKED — atomic, no Redis, safe with N workers. Each run executes in its own subprocess because the vendored engine uses process-global singletons (in-process multi-tenancy would be unsafe).
- BYOK decouples model economics (above).
- Observability from day one: Sentry on both frontend and worker, structured JSON logs with run IDs, and a health endpoint for the deploy platform.

The result: a production-grade, multi-tenant research platform whose monthly infrastructure cost is measured in small tens of dollars before scale — the margin structure an investor can underwrite.

5 · System Architecture

End-to-end view — from two human roles (customer, admin) through the React SPA, Supabase, the Python worker and the research engine, to DeepSeek and Paystack:

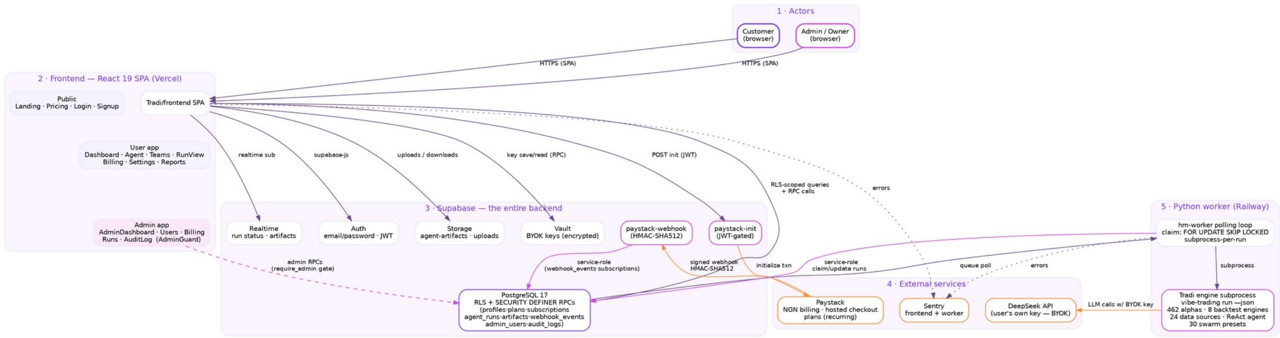

Figure 1 — Layered architecture. Dashed edges are admin-gated (SECURITY DEFINER RPCs with a require-admin gate); dotted edges are observability.

5.1 Component Map

Component	Role	Key facts
Frontend SPA	React 19 + Vite + TypeScript, Tailwind, Zustand	25+ pages; public / user / admin surfaces; AuthGuard; AdminGuard; realtime progress
Supabase Auth	Identity	Email/password, JWT, onAuthStateChange; branded email templates
PostgreSQL 17	Data + security	RLS on every table; SECURITY DEFINER RPCs with pinned search_path; billing + admin + audit tables
Supabase Storage	Artifacts	agent-artifacts and agent-uploads buckets
Supabase Vault	BYOK key encryption	vault.secrets; keys never stored in plaintext
Edge Functions	Paystack init + webhook	JWT-gated init; HMAC-SHA512 webhook with re-verification and idempotency
Python worker	Run execution (Railway)	Polling loop, FOR UPDATE SKIP LOCKED claim, one

		subprocess per run, structured logs, /health
Tradi engine (vendored)	Research engine	462 alphas, 8 backtest engines, 24 data sources, ReAct agent, 30 swarm presets
Paystack	NGN billing	Hosted checkout, recurring plans, signed webhooks
DeepSeek	LLM (customer's key)	BYOK — never proxied or marked up by the platform
Sentry	Observability	Frontend + worker error tracking; DSN fail-open

5.2 The Run Pipeline (core flow)

4. User launches a run (or swarm preset); start_agent_run RPC validates auth → active subscription → DeepSeek key configured → suspension gate.
5. Row inserted as queued with a unique idempotency key; realtime notifies the browser.
6. Worker polls agent_runs WHERE status='queued' with FOR UPDATE SKIP LOCKED and claims one row.
7. Worker spawns a subprocess: vibe-trading run -p "<prompt>" --json --no-rich; the engine decrypts the user's key server-side (service_role RPC) and calls DeepSeek.
8. Progress streams back (trace → progress columns); artifacts upload to Storage.
9. Run completes/fails; UI renders artifacts live via Supabase Realtime.

A run is a row, a claim, a subprocess and a stream — nothing more. That simplicity is the design's resilience.

6 · Design Flow — Admin and User

Two personas, one platform. Customers move through signup → billing → research; the admin (owner) operates the platform through a guarded dashboard whose every privileged action is a database-level, audited RPC.

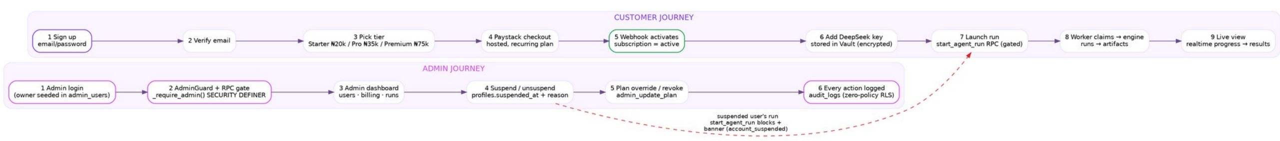

Figure 2 — Customer journey (top) and admin journey (bottom). The dashed red edge is enforcement: a suspended user's run is blocked at the database gate, not merely in the UI.

6.1 Customer flow

10. Sign up with email/password; verify via branded confirmation email.

11. Choose a tier; Paystack hosted checkout; the signed webhook activates the subscription (re-verified against Paystack's API, idempotent).
12. Add a DeepSeek key — encrypted into Vault via RPC (last-4 stored for display).
13. Launch single or swarm runs; watch live progress; read rendered research.
14. Billing callback page confirms state after checkout; renewals extend the period via webhook; failed renewals mark the account past_due.

6.2 Admin flow

15. Owner seeded in admin_users; /admin/* is guarded by AdminGuard and every RPC re-checks _require_admin() server-side (SECURITY DEFINER).
16. Dashboard surfaces stats, users, subscriptions, runs, audit trail.
17. Suspend/unsuspend sets profiles.suspended_at + reason; suspension blocks start_agent_run at the database gate and the UI shows the reason banner.
18. Plan overrides and revocations adjust subscriptions; every action is written to audit_logs (zero-policy RLS — append-only by construction).

Security boundary: admins never touch customer data through the client — only through audited, gated RPCs. Non-admins cannot read or write anything privileged even with a forged client.

7 · Functional Requirements

ID	Requirement	Where
FR-1	Email/password signup, login, logout, session persistence	Supabase Auth + Zustand store
FR-2	Email verification with branded templates, resend, expired-token state	Auth emails (dashboard-paste templates)
FR-3	BYOK key save / read / delete with last-4 display, encrypted at rest	Vault + RPCs (zero-policy user_api_keys)
FR-4	Launch single-agent and swarm (30 presets) runs with idempotency	start_agent_run / start_swarm_run RPCs
FR-5	Attachment uploads (CSV/XLSX/JSON, ≤50 MB, Premium)	Storage + uploads gate
FR-6	Live run status, progress streaming, artifact rendering	Realtime + progress columns
FR-7	NGN billing: checkout init, callback, subscription activation, renewals, past-	Edge Functions + webhook_events

	due, cancel	
FR-8	Admin: user list/detail, suspend/unsuspend, plan override/revoke, run and billing views	Admin RPCs + admin UI
FR-9	Audit trail of admin actions, append-only	audit_logs (zero-policy RLS)
FR-10	Monitoring: error tracking, structured logs, health endpoint	Sentry + logging_config + /health
FR-11	Pricing, landing, legal (ToS/Privacy with registrant block), support contact	Frontend pages + email templates

8 · Non-Functional Requirements

Category	Requirement	How met
Security	No client can read/write privileged data even when forged	RLS on every table; grants revoked from anon/authenticated where privileged; SECURITY DEFINER RPCs with pinned search_path
Security	Webhook authenticity	HMAC-SHA512 constant-time verification + server-side re-verification via Paystack API
Security	Secrets never in code or logs	Vault encryption; PII scrubber in logs; sk-redaction; DSN fail-open
Reliability	No lost payments or activations	Idempotency keys; 500-on-transient (retry) webhook semantics; duplicate suppression
Reliability	Safe multi-tenant execution	Subprocess-per-run isolation; FOR UPDATE SKIP LOCKED claim
Performance	Live progress perceived as instant	Realtime streaming; progress columns; 0/1 in-flight health proxy
Observability	Every run traceable	run_id threaded through JSON logs; Sentry on both apps
Cost	Near-zero variable cost per user	BYOK model; serverless Supabase; single small

		worker
Maintainability	Hermetic testability	53+ worker tests, validator-enforced design tokens, single source-of-truth CLAUDE.md

9 · Data & Security Model

Canonical schema (PostgreSQL 17). Every table has RLS; privileged tables have zero policies (default deny) and are reachable only through audited RPCs:

Table	Purpose	Access model
profiles	Public profile (id = auth.uid())	User-scoped RLS
plans	Tiers: starter / pro / premium, NGN price, Paystack plan code	Public read
subscriptions	Active plan, provider subscription id, period, status	User can SELECT own rows; writes via webhook/service role
agent_runs	Run queue + state machine	User-scoped; claims via gated RPCs
agent_artifacts	Result files	User-scoped
user_api_keys	BYOK metadata (last-4); secret lives in Vault	ZERO policies; RPC-only
webhook_events	Idempotency ledger (provider_event_id UNIQUE)	ZERO policies; service-role only
admin_users	Who may call admin RPCs	ZERO policies; _require_admin() gate
audit_logs	Append-only admin action log	ZERO policies; RLS on, no grants
usage_periods / usage_events	Historical usage analytics (pre-BYOK)	Read-only legacy

Security posture summary: JWT everywhere; admin = database-level gate, never UI-only; suspension enforced inside start_agent_run; billing money-movement fully webhook-driven and idempotent; keys encrypted in Vault with no plaintext path; logs scrubbed of secrets and PII.

10 · Project Phases & Current Status

Delivery record through a five-role pipeline (architect → coder → tester → reviewer → CEO) with automated verification on every phase:

Phase	Deliverable	Status

0–1	Auth, dashboard, agent launcher, run view	Shipped
2	Worker E2E (claim → subprocess → artifacts → Storage)	Shipped
3	BYOK pivot (Vault, RPCs, worker decrypt)	Shipped
4	Swarm (30 presets, Pro/Premium gate) + attachments	Shipped
5	Realtime queue, progress streaming, billing Edge Functions, callback, pricing, legal	Shipped
6	Admin dashboard (10 RPCs, suspension gate, audit trail)	Code shipped; migration APPLIED to production
7	Monitoring (Sentry, structured logs, /health)	Code shipped; DSNs created; wiring pending deploy
8	Paystack E2E (hermetic harness + webhook hardening, 2 bug fixes)	Shipped; live run pending test key
9	Email templates (branded, palette-enforced)	Shipped; dashboard paste pending
10	Cloudflare domain + SSL	Pending (manual)
11	Railway worker deploy (production env)	Pending (manual)
12	Launch prep (legal review, support mailbox, footer contact)	Pending

Testing discipline: 53+ hermetic worker tests, frontend production build gate, webhook signature verification tests, and a zero-dependency E2E harness for the billing flow. Deferred by design: live trading (mandate-gated, off), Stripe international billing (needs non-Nigerian entity), admin 2FA, public API, mobile views.

11 · Business Model & Unit Economics

- Revenue: flat monthly NGN subscription, prepaid via Paystack. No usage metering, no token markup.
- COGS: customer's own DeepSeek spend (outside the platform); infra is serverless + one small worker. Structurally >90% gross margin at scale.
- Acquisition: creator/community-led GTM (trading communities, educator channels) — the pattern that built African fintech user bases.

- Expansion: public API and mobile views open B2B2C channels (brokers, educators, communities reselling research); Stripe enables pan-African/international once entity structure permits.
- Risks acknowledged: pre-launch traction is zero today; CAC and churn are unproven — that is precisely what the GTM allocation underwrites.

"The engineering is done. The company is what the raise builds."

12 · Roadmap — What Funding Unlocks

Horizon	Deliverables	Enabled by
Launch (weeks)	Cloudflare + SSL, Railway production deploy, compliance review, support mailbox, public launch	Completion capital (small)
Growth (quarter)	Acquisition engine, referral program, community content, paid experiments; churn instrumentation	GTM allocation
Expansion (year)	Public API, mobile views, Stripe international, advanced analytics, admin 2FA; hiring	Product + working capital

13 · Key Architecture Decisions (ADRs)

ID	Decision	Rationale
D1	Subprocess-per-run	Engine uses process-global singletons; in-process multi-tenant is unsafe
D3	Reuse Tradi frontend (React, not Next.js)	Saves a rewrite; inherits the alpha UI
D5	SQL job claiming (FOR UPDATE SKIP LOCKED)	Atomic, no Redis, N workers safe
D8	Paystack + NGN only	Nigerian entity; Stripe cannot onboard
D11	BYOK pivot	Users supply keys; unlimited runs per tier; zero LLM margin risk
D12	Supabase Vault for key storage	Encryption without app-managed keys
D13	Edge Functions for Paystack	No FastAPI backend to operate
D14	No Celery/Redis — Python	Simpler; sufficient for MVP

	polling loop	scale

14 · Appendix — Repository Map & Testing

Path	Contents
Tradi/frontend/	React SPA — the production frontend (public + user + admin)
Tradi/	Vendored Vibe-Trading engine (MIT): agent, backtests, data sources, skills
vibe-trading-saas/worker/	Python worker: polling loop, runner, artifacts, logging, health
vibe-trading-saas/db/migrations/	SQL migrations (applied manually to Supabase)
supabase/functions/	Edge Functions: paystack-init, paystack-webhook
supabase/email-templates/	Branded auth + transactional email templates
scripts/	E2E harness + template validator (zero-dependency Node)
CLAUDE.md	Single source of truth: schema, RPCs, sprint tracker

Verification gates: worker pytest suite (53+ hermetic tests); frontend npm run build (type-check + bundle); webhook HMAC test vectors; billing E2E harness (offline fixtures + live mode with a hard sk_live_ guard); email template validator enforcing the design palette.

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